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Cybersecurity in the Era of generative and agentic AI: A reimagined architecture

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Abstract
The rapid adoption of machine learning (ML), generative artificial intelligence (GenAI), and agentic/hybrid AI systems demands a fundamental rethinking of cybersecurity strategy, architecture and governance. Traditional perimeter-and-identity based models are insufficient when autonomous agents act, learn and adapt inside enterprise systems. This paper (1) surveys unique threat surfaces introduced by ML/GenAI/agentic AI, (2) analyses gaps in current cybersecurity frameworks, (3) proposes an integrated defense architecture tuned for AI- native operation, and (4) outlines control, assurance and governance mechanisms to manage risk.

Keywords: Cybersecurity, machine learning, generative AI, autonomous agents, risk architecture, governance

1. Introduction
The disruptive rise of artificial intelligence systems has created novel opportunities and risks for enterprise and national security. Machine learning (ML) and generative AI (GenAI) are now embedded in business processes, automating decision-making, content/feature generation and system orchestration. Beyond these lies a growing class of agentic AI: autonomous or semi- autonomous software agents that perceive, plan, learn and act with minimal human oversight. (Cybersecurity Tribe)
From a cybersecurity perspective the implications are profound. Autonomous agents shift the identity/perimeter model: they become decision-makers, credentialed identities, and risk surfaces in themselves. (AI Frontiers)
Hence organizations must reimagine their cybersecurity posture: controls, architectures, governance, measurement and assurance cannot simply layer AI onto legacy frameworks. This paper outlines how.

2. Threat Surfaces & Attack Vectors Introduced by AI/ML/GenAI/Agents
A. ML model vulnerabilities
Adversarial inputs, poisoning, model extraction, membership inference—all classic ML risks— extend into AI-native systems. For example adversarial input may subvert a generative model producing malicious code or disinformation. (Wikipedia)

B. Generative AI misuse
GenAI can be weaponized: for phishing, social engineering, deep-fakes, automated malware generation. The defender's task expands from binary detection to nuanced behavior monitoring.

C. Agentic AI risk surfaces
Autonomous AI agents can (i) create or use credentials, (ii) act as independent identities, (iii) chain actions across systems, (iv) subvert human intent or bypass oversight. (AI Frontiers)

D. Infrastructure & supply-chain risk
AI platforms require massive data, compute and model supply chains. Risks include: corrupted training data, compromised model updates, GPU cluster vulnerabilities, third-party model mis- behavior, and multi-tenant cloud exposure.

E. Trust, identity and governance risks
When agents act autonomously the questions of accountability, auditability, explainability and compliance become acute. Traditional identity models (user, machine) must now include "agent identity" with its own threat modelling. (AI Frontiers)

3. Gaps in Traditional Cybersecurity Frameworks

A. Perimeter & identity-centric models
Most frameworks assume static identities (users, servers) and fixed boundaries. Autonomous agents erode those assumptions.

B. Static controls, limited adaptivity
Traditional rule-based detection struggles with polymorphic AI behaviors; models require dynamic, context-aware defenses.

C. Metrics & risk models inadequate
Metrics such as patch-cycle, number of incidents, mean-time-to-detect (MTTD) do not capture agent-based risk (such as self-propagating actions, model-based attacks).

D. Governance and oversight lag
Existing compliance frameworks treat AI as a tool, not as an active actor. Agentic AI demands new governance models: drift monitoring, behavior audits, chain-of-responsibility.

E. Supply-chain & third-party model risk under-addressed
Models may be procured from third parties, fine-tuned with proprietary data, deployed in production. The risk of trojaned models or corrupted training datasets is under-covered.

4. Proposed Architecture for AI-Native Cybersecurity

A. Identity & credentialing for agents
Define an explicit identity layer for AI agents: issue credentials, attributes, permissions, audit trails. Agents must authenticate, authorize and log like human actors.

B. Behavioral baselining and anomaly detection
Use ML/AI for defense: build behavioral models for agent actions (e.g., account creation, external call-outs, model invocation). Anomalies trigger containment or escalation.

C. Model-risk management (MRM)
Integrate model-centric risk: training data lineage, validation, drift monitoring, adversarial- resilience testing, and fallback/off-switch mechanisms.

D. Infrastructure control and segmentation
Segment compute, data, network environments for models and agents. Enforce least-privilege, micro-segmentation for AI compute clusters, audit GPU usage, isolate agent sandbox environments.

E. Real-time orchestration and playbooks
Enable rapid detection-response playbooks tailored for AI incidents (e.g., model exfiltration, prompt-injection, rogue agent behavior). Agents may help in response but must be governed.

F. Auditability, explainability, human-in-loop
Ensure AI decisions (especially those affecting security) are traceable, explainable, logged and subject to human review.

G. Lifecycle governance and controls
From procurement to deployment to retirement of models/agents. Controls include model- update review, versioning, rollback capability, performance/security monitoring.

5. Risk, Control & Assurance Framework

A. Risk modelling
Define risk scenarios specific to AI/ML/agents: e.g., a rogue agent exfiltrates data, or a generative model is exploited to generate zero-day code. Compute likelihood × impact.

B. Control mapping
Map controls to risks: e.g., for prompt injection attacks use input sanitization, limiting external context, least-privilege. For agent identity misuse use credential lifecycle, anomaly detection.

C. Assurance metrics
New metrics: number of agent identities created/retired; drift detected in model behavior; number of model updates failing adversarial testing; time to agent disablement; incidents involving agent-to-agent propagation.

D. Governance and oversight
Form cross-functional AI-risk board: include CISO, model owners, legal/compliance. Review agent deployments, drift, emergent behaviors, third-party model risk.

E. Continuous monitoring & control
Automated dashboards feeding into risk metrics, periodic red-teaming (including AI-agent team), and scenario-based simulations (e.g., rogue agent attack).

6. Implementation Challenges & Research Directions

A. Technical limitations
Explainability and interpretability of large models remain weak. Monitoring agent behavior at scale remains nascent.

B. Human-machine interaction
Determining when to intervene, how to limit autonomy, and how to architect human-in-the- loop versus full autonomy.

C. Supply-chain and third-party model risk
Need standards for model provenance, verification, tamper-resistance.

D. Governance, ethics and accountability
Who holds liability when a self-learning agent causes a breach? How to instill fairness, bias controls, alignment of incentives? (Res ea rchG ate)

E. Scale, cost and measurement
Large AI workloads consume resources; setting the business case for secure AI operations remains open.

F. Standardization and regulation
Need frameworks that cover agentic AI, model-based risk, identity for agents, emergent behaviors and cross-domain threat propagation.

7. Case Study & Illustrative Scenario

Consider an enterprise deploying a GenAI assistant to automate IT-service desk tasks, including account

provisioning and firewall rule changes. An agentic extension allows the assistant to act autonomously across systems.

Threat scenario: A malicious actor uses prompt-injection to escalate the agent's privileges and create new identities, then exfiltrates data via the agent's access path. Traditional controls (user MFA, network segmentation) fail because the agent is an authorized identity.

7.1 Mitigation architecture

- Agent identity onboarding process with credential binding and expiration.
- Behavioral monitoring of agent actions (e.g., unusual account-creation patterns).
- Input sanitization and sandboxing of prompt contexts.
- Model-update governance: before enabling autonomous behavior the assistant must pass adversarial testing.
- Incident playbook: disable agent, isolate environment, revoke credentials, restore from backup, audit logs.

Outcome: Enables safe deployment of the automation, while controlling the novel risk vector.

8. Conclusion

AI/ML/generative/agentic systems are redefining the threat landscape and stretching traditional cybersecurity paradigms. Organizations must re-engineer their architectures, processes, and governance to account for autonomous agents, model-driven risk, and dynamic behaviors. The architecture, controls and frameworks outlined in this paper provide a blueprint for reimagined cybersecurity aligned to the new era. Future research should focus on scalable monitoring, agent identity management, standardization of model-risk controls, and regulatory frameworks that evolve with technology.

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